Solution Architecture

Date	12-06-2026
Team ID	
Project Name	**Voice Based Concept Understanding Analyser**
Maximum Marks	4 Marks

Solution Architecture:

The Voice-Based Concept Understanding Analyser (VBCUA) is an AI-powered web application designed to evaluate a student's conceptual understanding through spoken responses. The system allows students to upload voice recordings via a Streamlit-based interface, after which the FastAPI backend processes the audio using OpenAI Whisper for speech-to-text conversion, Sentence-BERT for semantic similarity analysis, and Librosa for speech feature extraction. Based on the student's explanation and speech quality, the Google Gemini API generates personalised feedback highlighting strengths, weaknesses, and areas for improvement. The evaluation results, including scores and feedback, are stored in an SQLite/PostgreSQL database, and a detailed assessment report is generated in PDF format using ReportLab. By combining speech recognition, natural language processing, audio analysis, and generative AI, the system provides an efficient, objective, and intelligent alternative to traditional oral assessments.

Architecture Diagram: